

```

import cpmPy as cp
import math

def rel2str(rel):
    """ print congruence relation as nontrivial classes """
    result=''
    keys=sorted(rel.keys())
    for a in set(rel.values()):
        vs=[str(v) for v in keys if rel[v]==a]
        if len(vs)>1:
            result += ' '.join(vs)+'|'
    return result

def sol_key(rel):
    return (len(set(rel.values())),rel2str(rel))

def find_folding(Opt: dict[int, list[int]]) -> list[dict[int, int]]:
    """ find all folding relations of the optiongraph Opt """
    V=sorted(Opt.keys())
    k=len(V)
    mdl=cp.Model()

    # class number
    cl={p:cp.intvar(1,k) for p in V}

    # partition
    mdl+=cl[V[0]]==1
    for i in range(1,len(V)):
        mdl+= cl[V[i]] <= 1+cp.max([cl[V[j]] for j in range(i)])

    # arrow in the quotient rulegraph
    def Qar(A,B):
        return cp.all([(cl[a]==A).implies
                        (cp.any([cl[b]==B for b in Opt[a]]) for a in V)])

    # folding
    for a in V:
        for b in Opt[a]:
            mdl+=Qar(cl[a],cl[b]) | cp.any([cl[c]==cl[a] for c in Opt[b]])
            # mdl+=Qar(cl[a],cl[b]) | Qar(cl[b],cl[a])

    sols=[]

    def add_sol():
        result={p:cl[p].value() for p in V}
        sols.append(result)

    mdl.solveAll(display=add_sol)

```

```
sols.sort(key=sol_key)
return sols
```

Figure 5.1

```
Opt={1: [2], 2: [3, 4, 5], 3: [6], 4: [5], 5: [6], 6: []}
rels=find_folding(Opt)
for rel in rels:
    print(rel2str(rel))
```

```
|1 4 6|3 5|
|1 4 6|
|3 5|4 6|
|3 5|
|4 6|
|
```

Figure 5.2

```
Opt={1: [2, 3], 2: [4, 5], 3: [4, 5], 4: [], 5: []}
rels=find_folding(Opt)
for rel in rels:
    print(rel2str(rel))
```

```
|1 4 5|2 3|
|1 4 5|
|1 4|2 3|
|1 5|2 3|
|2 3|4 5|
|1 4|
|1 5|
|2 3|
|4 5|
|
```

Figure 5.3

```
Opt={1: [3], 2: [4], 3: [5], 4: [5], 5: []}
rels=find_folding(Opt)
for rel in rels:
    print(rel2str(rel))
```

```
|1 2 5|3 4|
|1 2 5|
```

```
|1 2|3 4|
|1 5|3 4|
|2 5|3 4|
|1 5|
|2 5|
|3 4|
|
```

Figure 5.4

```
Opt={6: [5], 5: [4], 4: [3], 3: [2], 2: [1], 1: []}
rels=find_folding(Opt)
for rel in rels:
    print(rel2str(rel))
```

```
|1 3 5|2 4 6|
|1 3 5|2 4|
|1 3 5|2 6|
|1 3 5|4 6|
|1 3 5|
|1 3|2 4|
|1 3|
|
```

Figure 9.1

```
Opt={1: [3, 4], 2: [4], 3: [5, 6], 4: [6], 5: [], 6: []}
rels=find_folding(Opt)
for rel in rels:
    print(rel2str(rel))
```

```
|1 2 5 6|3 4|
|1 2 5 6|
|1 2|3 4|5 6|
|1 5 6|3 4|
|2 5 6|3 4|
|1 2 6|
|1 5 6|
|2 5 6|
|3 4|5 6|
|1 6|
|2 6|
|5 6|
|
```

Example 11.2: Totative

```
Opt={i:[j for j in range(1,i) if math.gcd(i,j)==1] for i in range(1,10)}  
print(Opt)  
rels=find_folding(Opt)  
print(len(rels))  
for rel in rels:  
    print(rel2str(rel))
```

```
{1: [], 2: [1], 3: [1, 2], 4: [1, 3], 5: [1, 2, 3, 4], 6: [1, 5], 7: [1, 2, 3,  
4, 5, 6], 8: [1, 3, 5, 7], 9: [1, 2, 4, 5, 7, 8]}  
20  
|2 4 6 8|3 9|  
|2 4 6 8|  
|2 4 6|3 9| |
|2 4 8|3 9|  
|2 6 8|3 9|  
|2 6|3 9|4 8|  
|2 4 6|  
|2 4 8|  
|2 4|3 9|  
|2 6 8|  
|2 6|3 9|  
|2 6|4 8|  
|2 8|3 9|  
|3 9|4 8|  
|2 4|  
|2 6|  
|2 8|  
|3 9|  
|4 8|  
|
```